

Data Science – Univariate Analysis

1. Measure of central tendency

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0

Academic Scores Interpretation

SSC_P

- Mean, Median = **67**
- Most students scored around **62** (Mode)
- Consistent performance across students

HSC_P

- Mean **66.33**, Median - **65**, Mode - **63**
- Most students scored **63**, but a few high scorers pushed mean up

Degree Percentage

- Mean, Median, Mode – **65, 66**
- Most **stable and consistent** scores among all academic columns

Employability Test (etest_p)

- Mean - **72.1**, Median – **71**, Mode - **60**
- Majority scored around **60**, but top performers significantly raised the mean
- **Biggest performance gap** in the dataset

MBA Percentage

- Mean, Median - **62**
- Mode - **56.7** is lower

2. Percentile

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0

Academic Scores — Percentile Interpretation

SSC_P

Metric	Value
Q1	60.6
Median	67.0
Q3	75.7
Max	89.4

75% of students scored **below 75.7**

Top students reached **89.4**

HSC_P

Metric	Value
Q1	60.9
Median	65.0
Q3	73.0
Max	97.7

99th percentile = **91.86 - very** few scored above **90**

Degree Percentage

Metric	Value
Q1	61.0

Median	66.0
Q3	72.0
Max	91.0

Very consistent academic performance

Employability Test (etest_p)

Metric	Value
Q1	60.0
Median	71.0
Q3	83.5
Max	98.0

Top 1% scored up to **97%**

MBA Percentage

Metric	Value
Q1	57.945
Median	62.0
Q3	66.255
Max	77.89

3. Interquartile Range

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.334744	66.358558	72.100558	62.278186	277648.648649
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.129	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	91.15	88.5	98.0	77.89	390000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lesser	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Greater	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1	40.89	42.75	50.0	50.0	51.21	200000.0
Max	215	89.4	91.15	88.5	98.0	77.89	390000.0
kurtosis	-1.2	-0.60751	0.086901	-0.09749	-1.08858	-0.470723	-0.239837
skew	0.0	-0.132649	0.162611	0.204164	0.282308	0.313576	0.8067

var	3870.0	117.228377	112.063731	53.60471	176.251018	34.028376	2944596800.882517
std	62.209324	10.827205	10.586016	7.321524	13.275956	5.833385	54264.139179

ACADEMIC SCORES Analysis

SSC_P

Metric	Value
Mean	67.30
Median	67.0
Std Dev	10.83
IQR	15.1
Skew	-0.133
Kurtosis	-0.608

HSC_P

Metric	Value
Mean	66.33
Median	65.0
Std Dev	10.59
IQR	12.1
Skew	+0.163
Kurtosis	+0.087

Degree %

Metric	Value
Mean	66.36
Median	66.0
Std Dev	7.32
IQR	11.0
Skew	+0.204

Kurtosis	-0.097
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Employability Test (etest_p)

Metric	Value
Mean	72.10
Median	71.0
Std Dev	13.28
IQR	23.5
Skew	+0.282
Kurtosis	-1.089

MBA %

Metric	Value
Mean	62.28
Median	62.0
Std Dev	5.83
IQR	8.31
Skew	+0.314
Kurtosis	-0.471

Variance Values

Metric	Value
SSC_P	117.23
Hsc_p	112.06
Degree_p	53.60
etest_p	176.25
mba_p	34.03

77.89

77.89

